הפקולטה למדעי הטבע
המחלקה למדעי המחשב

אוניברסיטת בן-גוריון בנגב
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Project Objectives

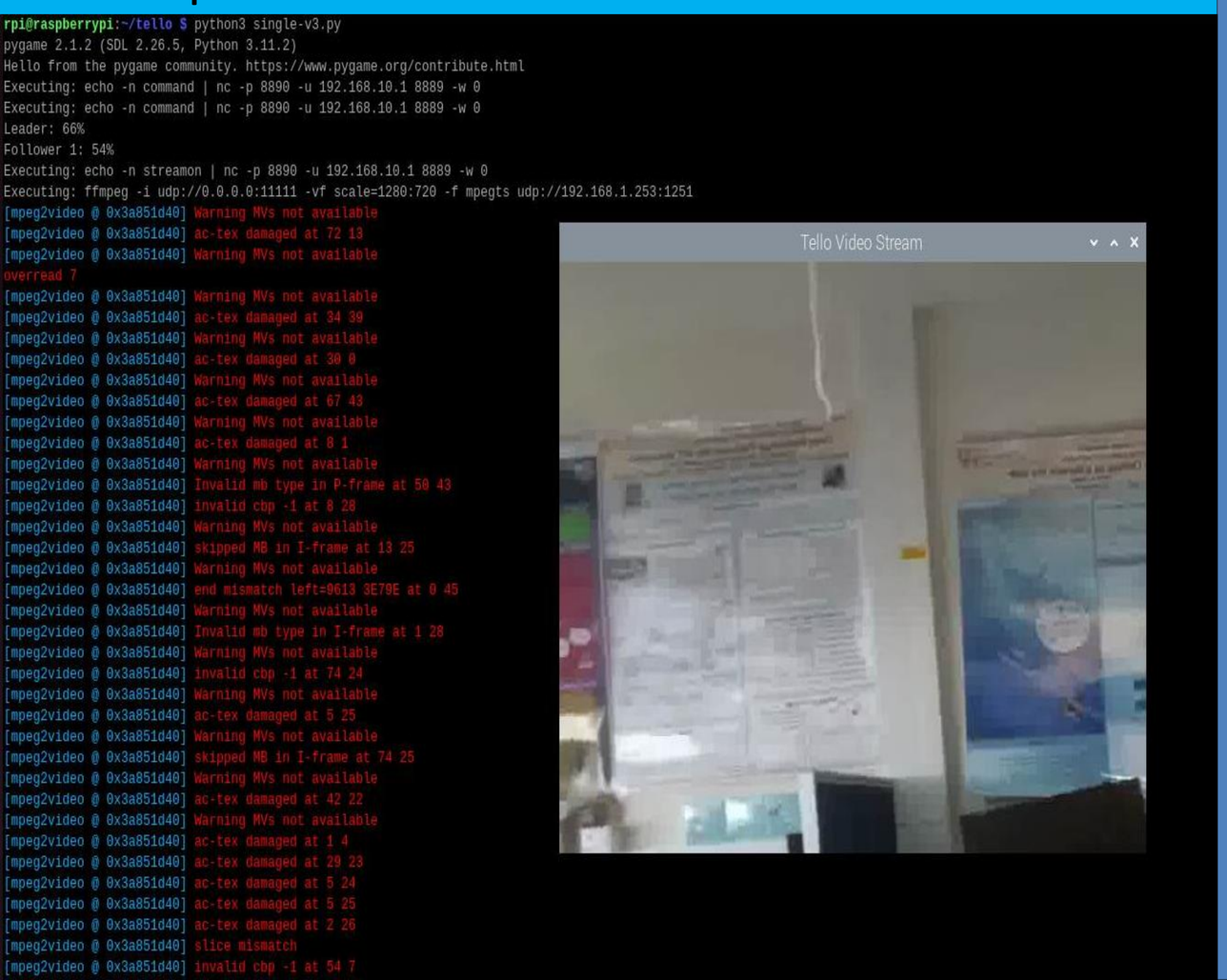
Design and implement a modular conversion kit, transforming off-the-shelf drones into autonomous assistants enabling seamless flight over vast areas.

Our Implementation

Mounting a Raspberry Pi and a wireless network adapter on each drone, enabling them to operate as network and battery range extenders, and allowing them to execute compound commands.

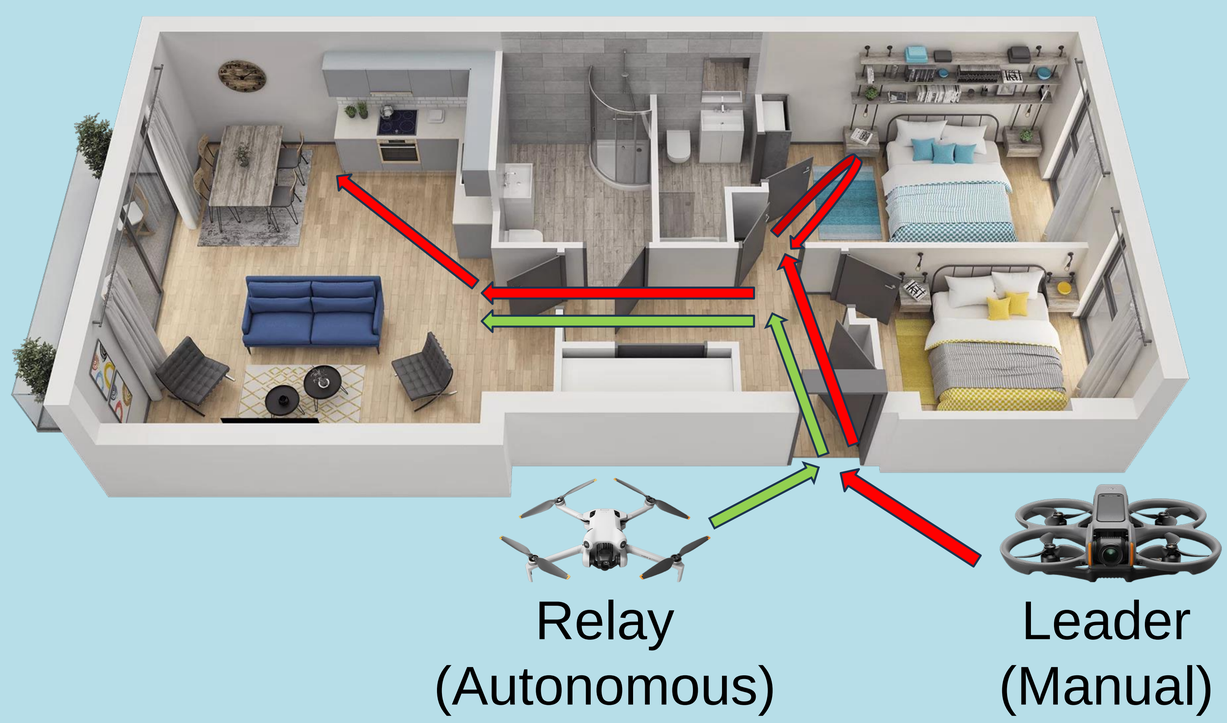
Control Software

Striving to make the interface as easy to master as possible, the drones (manual and autonomous) are controlled with intuitive WASD controls, enabling drone operators to focus on one drone at a time



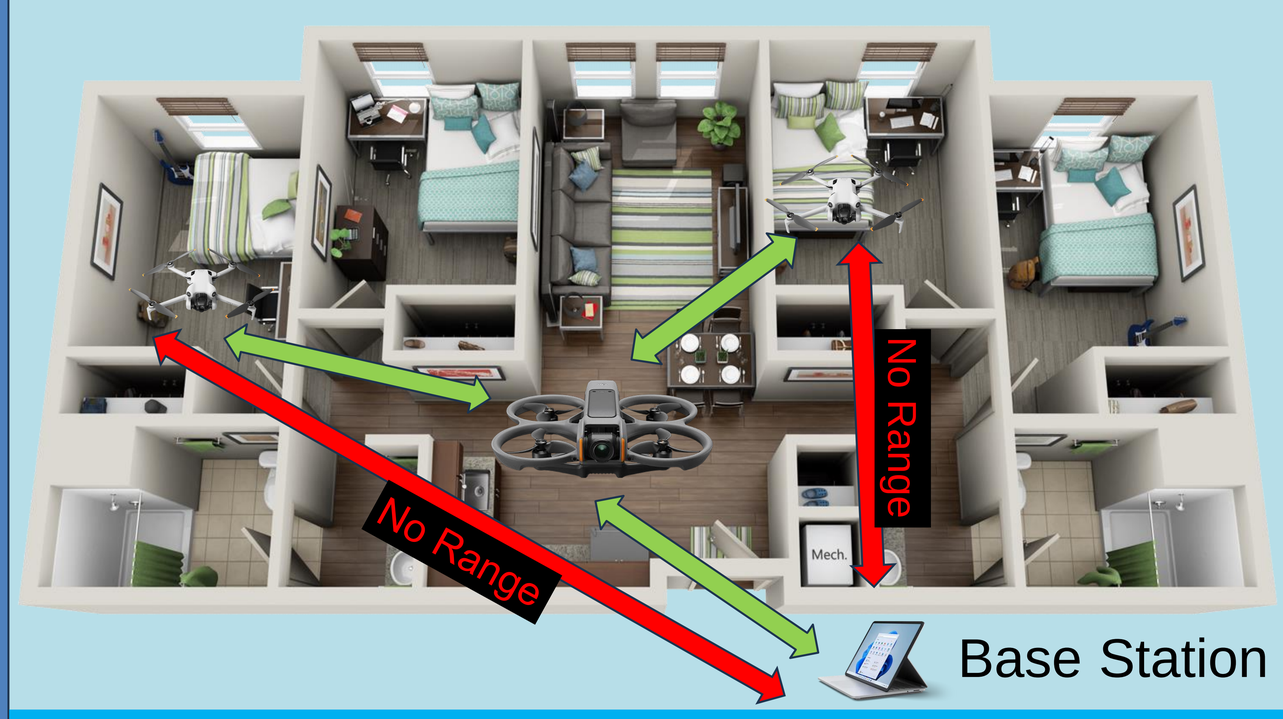
Autonomous Flight

One drone is controlled manually, and the rest autonomously extend its range and return to base when out of battery, following an optimized path.



Network Configuration

Raspberry Pis are nodes in a mesh network, upon which flight commands are reliably sent via the Raspberry Pis to the drones (TCP), and video is streamed back to the control software (UDP).



Physical Specification

Mounted on each drone is a Raspberry Pi Zero 2W with a TP-Link TL-WN725N network adapter, at a total cost of less than 25$ and a total weight of less than 15g. The kit was designed with the DJI Tello drone in mind but can be easily adapted to any others supporting rudimentary flight commands.